

The Chemistry of Human-Caused Climate Change

CHEMISTRY · CLASSES 9–10 + O/A-LEVEL EXTENSION · 70–80 MIN (DOUBLE PERIOD)

Prepared by Babar Ali — Science / CS Teacher

Learning Objectives

- Write and balance combustion equations for hydrocarbon fuels.
- Explain, at the molecular level, why CO_2 , CH_4 and N_2O absorb infrared radiation while N_2 and O_2 do not.
- Identify the main anthropogenic sources of the three key greenhouse gases.
- (*Extension*) Relate radiative forcing to concentration, absorption strength, and atmospheric lifetime.
- (*Extension*) Write the ocean-acidification equilibrium and explain its effect on marine carbonate chemistry.

Prior Knowledge Required

- Basic combustion reactions and balancing equations (Class 9 level).
- Simple molecular geometry / shapes of molecules (helpful for the IR-activity explanation).
- Concept of equilibrium (needed only for the acidification extension section).

Lesson Structure

Time	Activity
10 min	Hook: show CO_2 concentration graph (Keeling Curve) — ask students to hypothesise the cause.
15 min	Direct instruction: combustion chemistry of fossil fuels (Section A).
20 min	Direct instruction: molecular basis of the greenhouse effect — IR activity and dipole moments (Section B).
15 min	Direct instruction: methane, nitrous oxide, and radiative forcing comparison (Section C).
10 min	(<i>Extension classes</i>) Ocean acidification equilibrium (Section D).
Remaining	Worksheet + discussion / wrap-up.

Section A — Combustion Chemistry

Fossil fuels are hydrocarbons. Complete combustion of an alkane follows the general pattern:



Worked example (methane, the main component of natural gas):



Teaching point: every carbon atom that was locked underground for millions of years is oxidised to CO_2 and released into the atmosphere. Natural removal processes (photosynthesis, ocean/sediment uptake) work far more slowly than the rate at which humans are burning fuel — this imbalance is the core of the problem.

Section B — Why CO_2 Traps Heat (Molecular Basis)

This is the key conceptual section. CO_2 is a linear molecule ($\text{O}=\text{C}=\text{O}$) with three vibrational modes: symmetric stretch, asymmetric stretch, and bending.

- The symmetric stretch does **NOT** change the molecule's dipole moment $\rightarrow$ it is IR-inactive.
- The asymmetric stretch and the bending mode **DO** change the dipole moment $\rightarrow$ these are IR-active.

A vibrational mode can only absorb infrared radiation if it changes the molecule's dipole moment. Earth re-emits absorbed solar energy as longwave infrared radiation (heat). CO_2 's IR-active modes absorb this radiation (strongly around $15\mu\text{m}$) and re-emit it in all directions, including back down to the surface — this is the greenhouse effect at the molecular level.

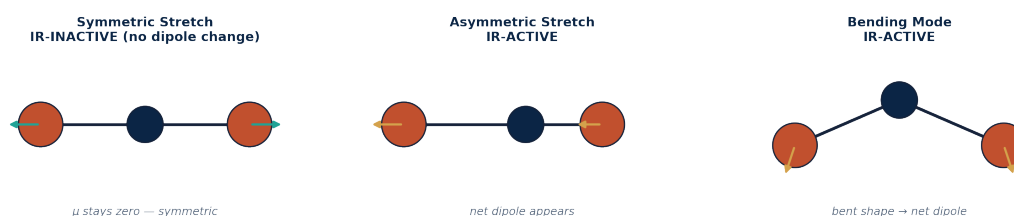


Figure 1. The three vibrational modes of CO_2 . Only the modes that change the dipole moment can absorb infrared radiation.

Contrast: N_2 and O_2 make up about 98% of the atmosphere but are diatomic and symmetric. Any vibration in a symmetric diatomic molecule cannot create a dipole moment, so these gases are IR-inactive and do not contribute to the greenhouse effect.

Section C — Methane, Nitrous Oxide & Radiative Forcing

Methane (CH₄)

Tetrahedral geometry; its stretching and bending modes are also IR-active. Per molecule, methane traps roughly 25–80× more heat than CO₂ (timeframe-dependent), but has a shorter atmospheric lifetime (~12 years vs. centuries for CO₂). Main sources: enteric fermentation in livestock, rice paddies, landfills, and leaked natural gas.

Nitrous Oxide (N₂O)

Produced mainly from nitrogen fertiliser via microbial processes:

Nitrification: $\text{NH}_4^+ \rightarrow \text{NO}_3^-$ (N₂O released as a byproduct)

Denitrification: $\text{NO}_3^- \rightarrow \text{N}_2\text{O} \rightarrow \text{N}_2$ (incomplete reduction leaks N₂O)

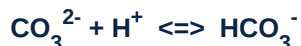
N₂O is roughly 300× more potent than CO₂ per molecule and persists ~114 years in the atmosphere.

Radiative Forcing — why CO₂ still dominates

A gas's overall warming contribution (measured in W/m²) depends on three factors together: how strongly it absorbs IR, its atmospheric concentration, and its lifetime. CO₂ dominates total impact not because it's the most potent per molecule, but because of its sheer atmospheric concentration (~420 ppm vs. ~1.9 ppm for methane).

Section D — Ocean Acidification (*Extension / A-Level*)

About a quarter of emitted CO₂ dissolves into the oceans, shifting a carbonate equilibrium:



The extra H⁺ lowers ocean pH and consumes carbonate ions (CO₃²⁻), reducing the material available for marine organisms to build calcium carbonate shells and skeletons.

Student Worksheet

- 1 Balance the combustion equation for propane (C₃H₈) burning completely in oxygen.
- 2 Explain in your own words why N₂ and O₂ do not contribute to the greenhouse effect, but CO₂ does.
- 3 Name the two main microbial processes that release N₂O, and give one agricultural source.
- 4 Methane is more potent than CO₂ per molecule, yet CO₂ causes more total warming. Explain why, using the idea of radiative forcing.
- 5 (*Extension*) Write the equilibrium showing how dissolved CO₂ lowers ocean pH, and explain the effect on marine organisms.

Answer Key (Teacher Copy)

- 1 $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$
- 2 N_2 and O_2 are symmetric diatomic molecules — their vibrations cannot change the dipole moment, so they cannot absorb IR radiation. CO_2 's bending and asymmetric stretch modes do change its dipole moment, making it IR-active.
- 3 Nitrification and denitrification; nitrogen-based fertiliser use is the main agricultural source.
- 4 Radiative forcing depends on absorption strength, concentration, AND lifetime combined. CO_2 's far higher atmospheric concentration (~420 ppm vs ~1.9 ppm for CH_4) outweighs methane's higher potency per molecule, giving CO_2 greater total forcing.
- 5 $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3 \rightleftharpoons \text{H}^+ + \text{HCO}_3^-$ (and $\text{CO}_3^{2-} + \text{H}^+ \rightleftharpoons \text{HCO}_3^-$). Lower pH and reduced CO_3^{2-} availability make it harder for marine organisms to form calcium carbonate shells/skeletons.

Suggested Follow-Up

- Show a real Keeling Curve dataset and have students plot/interpret the trend (cross-link to Computer Science / data handling).
- Compare radiative forcing values for CO_2 , CH_4 , and N_2O using IPCC data as a short research task.
- Extension discussion: link chemistry content to Biology (carbon cycle) for a cross-curricular angle.